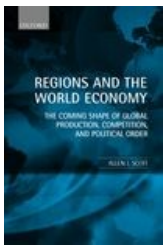

Regions and the World Economy: The Coming Shape of Global Production, Competition, and Political Order

Allen J. Scott

https://doi.org/10.1093/acprof:oso/9780198296584.001.0001

Published: 16 December 1999 **Online ISBN:** 9780191685255 **Print ISBN:** 9780198296584

CHAPTER

5

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The Regional Foundations of Economic Performance

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https://doi.org/10.1093/acprof:oso/9780198296584.003.0005 Pages 74–100

Published: December 1999

Abstract

To further understand the network of regional economies, this chapter explains economic 'performance' through looking into growth and development manifested in variables such as productivity, employment, competitiveness, innovation, and others. The author attempts to develop a location view that considers the dependence of economic order on geographical aspects. The author introduces the notion of an *economic space* that can assume various forms in characterising any social division of labor. Externalities that cause either benefits or costs on producers are also discussed in relation to economic space. The chapter also analyses several theories that may have influenced the author's concept of a location theory and spatial agglomeration theory.

Keywords: economic performance, location, economic space, division of labor, agglomeration

Subject: Political Economy, International Business

Collection: Oxford Scholarship Online

In previous chapters I have tried to lay out a broad empirical panorama of the rise and nascent consolidation of a globally interlinked network of regional economies. In order to push our understanding of these matters further forward, we now must shift into a more conceptual mode of investigation and look carefully at some basic ideas about the relations between geography and economic performance, where by 'performance' I mean growth and development as manifest in such specific variables as productivity, innovation, employment, competitiveness, and so on.

The point here is that we need both to clarify the locational logic of productive activity and to show how this logic then operates in reflexive relation to the dynamics of performance. This task involves much more than a simple rehearsal of classical location theory, which in its received versions (i.e. accounts of the spatial geometry of the economy) has always been a rather minor appendage to economic theory as a whole. The task before us here is to develop a view of location that is also a set of propositions about the necessary dependence of the economic order at large on geography. This is a task that modern economic geographers have begun to tackle

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with much success, and with some dramatic consequences, too, for the ways in which we can begin to think about local economic development policy.[1]

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Industrial Organization and Economic Space

The structure of economic activity

There is a long tradition of economic analysis, ranging from Böhm-Bawerck[2] through Young[3] and Leontieff[4] to Perroux,[5] Isard,[6] and beyond, in which production is described as a social division of labor whose discrete nodes (i.e. individual producers or aggregates of producers) are held functionally together by extensive networks of transactions. In such a system, any producer, say A, sells a quantity of output to another producer, say B; B adds some value to the product and then sells to C; C adds further value and sells to D; and so on, until at some stage the product is taken out of the circuit of production and consumed. At any point in the circuit, some of the product can return to an earlier node for further processing; then, as the division of labor deepens and widens, we say, following Böhm-Bawerck, that the *roundaboutness* of the circuit is increasing, i.e. that the number of processing stages between raw-material inputs and the final consumer is expanding.

This descriptive model can be immediately enriched by thinking of each node in the circuit of production as an input—output mechanism such that there is a definite relation between a quantity of given input (e.g. sheet metal or plastic parts) and a quantity of derivative output (e.g. aircraft or computers). A basic condition sustaining the existence of any pre-defined division of labor is that producers can always purchase their inputs more cheaply and effectively from outside sources than they can make them themselves.

The latter remark takes us at once into questions of industrial organization as originally formulated by Coase in 1937 and Williamson in 1975, and which provide some important foundations for the integration of issues of
p. 77 economic location and performance.[7] In their ↳ essence, these questions boil down to a series of theoretical propositions about the relations between firms and markets, and above all about their *scope*, i.e. the range of interconnected types of production tasks internalized within any one system of managerial control (the domain of the firm) versus the range and types of interconnected firms that make up the social division of labor (the domain of the market). When the internal scope of firms is increasing (which in a static world means that the external scope of the market is decreasing), we say that *vertical integration* is occurring. An example of this phenomenon is the combination of semiconductor production and computer manufacturing into single units of production where there are strong technological interdependencies between the two, e.g. where their common internalization allows the free flow of technological information and know-how between them, thereby enhancing their overall efficacy. If, by contrast, firms are shedding functions and their internal scope is decreasing (with the external scope of the market increasing as a consequence), we say that *vertical disintegration* is occurring. An example here might be the splitting up of cotton-spinning and weaving functions into separate firms thus allowing them to attain higher levels of production flexibility as represented by more varied outputs or more frequent design changes. It should be noted that between the domains of the firm and the market there are many possible intermediate (quasi-internal, quasi-external) organizational states including, for example, informal agreements, written contracts, joint ventures, partnerships, *keiretsu*-style associations, and so on, that add a further layer of complexity to the structural fabric of production.[8]

Coase and Williamson, together with a number of other industrial organization theorists,[9] have provided elaborate analytical accounts of the many and varied conditions that lead to vertical integration or
p. 78 disintegration of production systems. It is not necessary here to dwell on ↳ the details of these accounts except to observe that a calculus of transactions costs tends to play a significant role in their analytical machinery. We may observe too, following Granovetter and others, that a strictly economic logic of production will take us only so far in understanding industrial organization processes, and that we need also to take into

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account the fact that transactional systems are always and of necessity embedded In historically determinate social conditions.[10] Thus, in the example of semiconductor production and computer manufacturing cited above, it is possible that vertical integration might be deferred or only partially implemented if social conditions were such as to promote high levels of trust between firms, thus permitting requisite flows of information to occur without formal amalgamation. In any case, a given economic system will always be constituted as a complex network of relatively integrated and relatively disintegrated firms in a constant state of adjustment and readjustment as shifts in technologies, labor markets, the competitive environment, and so on occur.

Any social division of labor (whether defined in terms of sectors or firms) can be thought of as defining an *economic space* in which every firm or sector, as the case may be, has an explicit input—output relationship to all other firms or sectors. We must distinguish this notion of economic space from the notion of *geographic space*, which represents a locus of actual places, whether functionally interrelated with one another or not. Economic space itself can assume many different structural forms, though two archetypes are of some relevance to the present discussion. One of these is represented by pyramid-like industrial complexes where large lead plants (often possessing a wide scope of internalized production activities) sit at the top of transactional hierarchies of smaller input suppliers, service providers, and subcontractors (see Figure 5.1). This polarized organizational form is typical of the car or aerospace industries. The other involves finely grained transactional networks linking together many small and vertically disintegrated producers, without the strong centralization effects that are to be found in the former case. This type is exemplified by industries such as clothing, jewelry, or advanced semiconductor manufacturing (see Figure 5.2). Many different sorts of hybrid or
p. 79 amorphous expressions of economic space are possible too. Whatever their specific ↳

p. 80 ↳

p. 81 ↳
geometric outlines, they all represent structures of economic interdependence held together by an organic social division of labor.

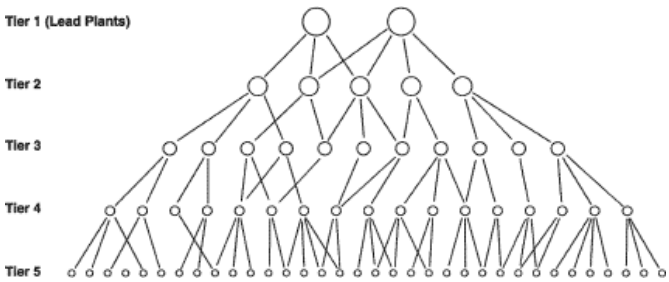

Fig. 5.1. A pyramid-like economic space

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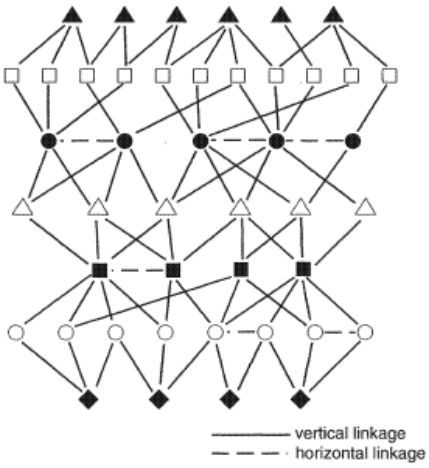

Fig. 5.2. A diffuse economic space with vertical and horizontal linkages

Externalities

By the same token, economic space is also the site of numerous externalities which impose definite benefits and costs on individual producers. An externality is any occurrence or activity that lies outside the range of control of the individual firm, but that then has definite effects on the firm's internal production function. Some externalities are internalized by firms through normal market channels, as for example when a firm buys inputs from another firm that can produce them to a higher standard of tolerance than it can itself achieve, Other externalities—and these are among the most interesting ones from our standpoint—are prone to market failures of various kinds and thus are not subject to the logic of the pricing system. These externalities may be either positive (e.g. information spill-overs that supply useful knowledge *gratis* to firms that did not participate in its creation) or negative (e.g. air pollution, or cheap imitators of a fashion trend that then becomes quickly debased). Many kinds of externalities exhibit increasing returns effects. This means that the more abundant they are, the more beneficial in proportional terms they become (if positive) or detrimental (if negative) to the firms that internalize them.

Of all the many and various externalities that exist in any production system, some of the most potent derive from the fact that economic space is by definition a domain of exchange, interaction, and flow. It can be represented not only as a field of physical input-output transactions between producers, but also as a gigantic never-ending multi-way discussion in which individual economic agents exchange, barter, deal, dispense instructions, learn from one another, come to know and to understand one another, and in general keep the wheels of commerce fully greased. It is, in short, a social and interpersonal as much as an economic network in the narrow sense.

For example, the innovative capacities of individual producers are frequently much enhanced by the bits and pieces of information that circulate informally and continuously through economic space. Significant elements of this kind of information, of course, represent so much floating chaff, but some of it has surprisingly
p. 82 stimulating and beneficial repercussions on economic activity. Thus, as both von Hippel ↳ and Russo have indicated, a dialogue between a supplier and a buyer about a particular order and its specifications—especially where mismatches between the supplier's initial capabilities and the buyer's initial demands become apparent —can often lead to innovative insights, e.g. on the one side about how process configurations might be improved, and on the other about how alternative product designs might be put more effectively to work.[11] This type of innovation almost always entails small-scale, informal, day-by-day refinements of processes and products, and it needs to be distinguished from the kinds of innovation that come from large-scale research efforts associated with aprioristic projects like the quest for space satellites, supercomputers, effective batteries

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for electric cars, and so on. Nevertheless, the cumulative effect of these small refinements over time can often be enormous, and will usually be most intense where economic interdependencies are strongly developed.

Another major type of externality that is closely dependent on the transactional structure of production concerns the formation of peculiar cultural niches in certain articulations of economic space. In view of the high levels of interaction and interdependence among groups of individuals caught up in particular constellations of productive activity, they often acquire distinctive social perceptions, understandings, attitudes, and even vocabularies. Specific styles of competition and collaboration, tacit social agreements between different parties (e.g. management and labor), common frames of reference for judging reputations or trustworthiness—all are susceptible to construction in this manner. In many instances, as Salais and Storper have written, we can think of these phenomena as conventions, for even when they are not explicitly formalized, they are often represented by indurated habits, rituals, and ideologies that guide human action in predictable directions and that have forceful impacts on business operations in both positive and negative ways.[12] Segments of the economy such as electronics manufacturing, financial services, or the film industry
p. 83 are all heavily marked by traits of this sort, and we can properly think ↳ of these segments as distinctive cultural milieux, though the reader must again be reminded that the discussion is still focused on economic and not geographic space. Indeed, Marshall's celebrated though by now heavily overworked dictum about 'industrial atmosphere' and its constitution as an overarching structure of useful customs and sensitivities[13] can be interpreted in the first instance and in notional terms as a placeless and strictly sectoral phenomenon adhering to purely functional communities of producers.

The spatiality of transactions

This description of the structure of economic space and the externalities that are interwoven with it highlights the critical importance of transactions in the everyday functioning of production systems. Transactions are in part an outcome of the division of labor and the resulting inter-firm flows of inputs and outputs. But they also function as channels of social communication, and, as just demonstrated, they are one of the bases of reproducible forms of cultural solidarity.

At the same time, transactions have a dual dimensionality. They exist not only in economic space but in geographic space as well. They are therefore marked by attributes of location and distance, meaning that they are stretched out between sets of origins and destinations, and that they accordingly incur various costs as the space between these locations is spanned. This signifies in turn that mutual locational proximity (i.e. in geographic space) of firms that are interlinked in economic space is often highly beneficial to all parties. The degree of benefit, however, is intimately dependent on a number of critical qualitative features of the transactional system, as indicated in the following three points.

First, very small-scale transactions, in which the value of what is being transacted is low *relative* to its spatially dependent costs (e.g. a small-batch order from a clothing manufacturer to a sewing subcontractor), will in general be economical only over short distances. Larger transactions enjoy economies of scale that make it possible to stretch them out over much longer distances.

Second, transactions that are continually changing in their specifications and are irregular in space and time
p. 84 are likely to be associated ↳ with high space-dependent costs. More standardized, streamlined, and predictable transactions can take advantage of significant savings in ordering, scheduling, packaging, investment in dedicated transportation facilities, and so on, thus allowing for more spatially extensive links. Firms that depend on irregular kinds of transaction often locate in dense and multifaceted agglomerations because the external economies of scope in such areas provide insurance against unpredictable eventualities.

Third, different modes of transacting have very different implications for spatially dependent costs. For example, face-to-face encounters, where two or more individuals must come together at a single location in

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order to deliberate over some exchange, are typically very costly. By contrast, the electronic transmission of messages is easily accomplished over even intercontinental distances.

A rough corollary of these three points is that firms whose transactions with one another are small in scale, irregular and unpredictable, and dependent on intensive face-to-face intervention will probably find it to their advantage to be located in some sort of mutual proximity, whereas firms whose transactions with one another have the opposite characteristics are likely to be more free in their choices of location.

Critics of the transactions-cost approach to location theory have sometimes claimed that it is too narrowly focused on the monetary costs attached to the physical transportation of finished and semifinished outputs. In actuality, spatially dependent transactions costs refer to any kind of geographic impediment to any kind of interaction (from the most concrete to the most insubstantial) between locations, including purely social and psychological evaluations of interlinkage. The point, however, is that space (or distance) is always fundamentally present as a barrier to be crossed. Even the purely monetary costs of transacting are highly multidimensional. Thus, in addition to being measured on a simple direct transport cost scale, they may also be measured in terms of personnel travel time, rates of deterioration of materials or information, velocities of circulation of inputs and outputs, the efficacity of just-in-time production systems, and so on. Moreover, these costs, in the widest sense of the term, apply to both traded and untraded forms of interdependency between individual transactors, whether we are talking about the transmission of train-loads of coal or about social
p. 85 encounters over lunch. Even in a world ↳ of advanced electronic communications technologies enabling some kinds of information to flow across the globe virtually without any time lag and at trivial monetary expense, geographic space still exerts major restraints on many kinds of transacting. This state of affairs will continue until such time as we have the equivalent of magic carpets, which is to say instantaneous and costless transmission of all goods and services (including labor services) in any quantity from any point in the world to any other point, on command.

The General Theory of Location

These remarks take us at once into the central questions of modern location theory, that is how and why different types of production occur at different levels of quantitative resolution in different places, and how any specific locational outcome affects the performative qualities of the economy.

Classical location theory has a long and intellectually rich history, ranging from the early formulations of Alfred Weber about cost-minimizing strategies for locating individual plants, through the central-place constructs of Christaller and Lösch, to the general spatial equilibrium analytics of the regional science school in the 1960s and 1970s.[14] Much of classical location theory examines the effects of physical transportation costs on individual locational decision-making and behavior, and then seeks to calculate the resulting geometric layout of the economic landscape. More recently, theorists of the so-called 'new economic geography' approach have sought to re-express location theory in terms of what we might call a historical-structural agenda. In this agenda (to be developed more fully later), the rationality of individual decision-making and behavior is subsumed within the analysis of locational patterns as the expression of spatial and temporal
p. 86 interdependencies in which collective effects like agglomeration, ↳ institutional frameworks, and system-wide path dependency are allowed their Ml play.[15]

We can begin the task of developing these ideas more folly (but without abandoning completely certain critical insights of classical location theory) by looking at the interrelations between transactions costs, externalities, and location under a series of controlled conceptual conditions. In this exercise, I shall restrict the notion of transactions costs to spatially dependent phenomena, i.e. costs that vary solely as distance varies; and I shall deal only with those kinds of *unpriced* externalities that are generated by the co-existence of multiple firms. Recall that such externalities often engender marked increasing returns effects.

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If we combine these two variables in various degrees of intensity, we observe a series of theoretical locational results as presented in Table 5.1. Notice that the table has one more column than rows in order to accommodate the special and extremely important case where spatially dependent transactions costs are heterogeneous in the sense that they are low for some kinds of interaction and high for others.

Table 5.1. Schematic locational outcomes resulting from the combination of spatially dependent transactions costs and externalities

Externalities	Spatially dependent transactions costs		
	Uniformly low	**Heterogeneous**	**Uniformly high**
Low	1. Spatial entropy	2. Random dispersal combined with emerging Löscherian-Weberian landscapes	3. Löscherian-Weberian landscapes
High	4. Small interconnected clusters	5. Super-clusters	6. Small disconnected clusters

We may examine the contents of the table by looking at each of its numbered panels in turn. The reader is warned that the following discussion is intended to be illustrative rather than analytically definitive, and it is in particular abstracted away from any historical context or developmental logic. Moreover, it should be kept in mind that transactional relationships and externalities are not, strictly speaking, independent of one another, for the former commonly provide channels through which access to the latter is obtained (as in the celebrated example of the apple orchard that supplies nectar to bees, while the bees pollinate the apple blossoms; however, the bees must transport themselves from the hive to the orchard and back again).

1. *Low transactions costs, low externalities.* As spatially dependent transactions costs fall, we approach closer
p. 87 and closer, to a state of the world represented by the magic-carpets analogy; and with ↳

p. 88 ↳
falling externalities (as defined), firms' performance levels become increasingly independent of one another. In the theoretically limiting case, where all transactions costs are absolutely zero (and even where positive externalities are present), Ml spatial entropy, i.e. locational randomness, will prevail.

2. *Heterogeneous transactions costs, low externalities*. In situations where some transactions costs are low and others high while externalities remain at a consistently low level, we are likely to encounter a hybrid situation represented by some random dispersal of locations combined with emerging Löschian-Weberian landscapes (see next paragraph).

3. *High transactions costs, low externalities*. Under these conditions, locational patterns will be uniformly and responsively structured by the effects of distance. This suggests that producers will tend to seek out locations that reduce transport costs on inputs and outputs to the lowest possible level, and/or provide optimal access to markets over a given spatial range. Since externalities are negligible, firms will be locationally indifferent to one another except insofar as they are connected by input-output relations. The concrete geographical structures that are apt to ensue may be described as either (*a*) Löschian central-place systems, or (*b*) Weberian landscapes characterized by an irregular but cost-minimizing distribution of producers reflecting underlying patterns of resource availability, input-output linkages, and the locations of final markets.

4. *Low transactions costs, high externalities*. As we shift from panel 1 in Table 5.1 to panel 4 (i.e. from low to

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high externalities in the context of uniformly low transactions costs), we are likely to begin to observe the formation of small production clusters made up of firms that generate particularly intense positive externalities for one another. These clusters will remain small since the low transactions costs make it possible for inter-firm linkages and externality effects to occur over long distances, thus eliminating the need for proximity except on a very selective basis. Under these circumstances, for example, one might imagine parts of the Hollywood film industry breaking up into geographically detached but strongly interacting sub-clusters each based on a specialized trade such as scenario writing, special effects, animation, photographic processing, film shooting, and so on.

5. *Heterogeneous transactions costs, high externalities.* This case is actually the most crucial one for the
p. 89
purposes of the discussion in ↳ this book, and I shall argue that the current moment of economic geography and history is one that is precisely described by the boundary conditions here, namely (*a*) spatially dependent transactions costs that range, for example, from the minuscule for international currency exchanges to onerous for many kinds of face-to-face encounters, and (*b*) high externalities, especially in the case of post-fordist flexible production systems. In the latter kinds of systems, most notably those marked by advanced levels of vertical disintegration, producers frequently cluster tightly together, both because a significant proportion of their mutual transactions incur high costs, and because the externalities that they jointly engender are copious. Yet at the same time, low costs on selected extra-local transactions enable each cluster to tap into world-wide resources and markets, thus stimulating massive local growth. Under the right combination of circumstances, the result can be the emergence of a large number of different super-clusters of capital and labor.

6. *High transactions costs, high externalities*. The solution proposed in the final panel of Table 5.1 is similar to that of panel 4, in that the prevailing locational pattern consists of small clusters, but with the difference that they are here relatively disconnected from one another. Actually, under the severe—indeed, strictly imaginary—analytical assumptions of a world that has no prior history and no geographic variations in basic resource availability, the trend would now be toward the emergence of one large agglomeration of producers. A more probable outcome, however, would be the case where we observe many small clusters, each with well-developed intra-cluster externalities, and each being located in relation to some antecedent condition of development or resource availability (as in the example of pre-capitalist craft communities). However, the high transactions costs will make it economically difficult for firms to engage with each other over extended distances, and this will therefore impose strict limits on any one cluster's ability to grow.

These schematic comments provide us with a rough preliminary sense of the potential complexity of locational outcomes, even when the basic framework conditions governing them are limited to just a few variations in transactions costs and externalities. Needless to say, this complexity is enormously increased if we append
p. 90
other conditions to the framework in order to take account of wider historical, social, and ↳ cultural forces. The argument that is in course of development in these chapters advances the thesis that in the current conjuncture the peculiar substantive thrust of the latter forces greatly intensifies the tendency for super-clusters to emerge. In this conjuncture, very significant elements of the production system are concentrated in locational magnets which draw their centripetal vigor from their abundant externalities combined with selectively high transactions costs; yet simultaneously, many of the goods and services that they produce flow with ease across the face of the earth. It is in this sense that we can speak of locational constraints on the globalization of production systems, even while the globalization of markets proceeds apace around us.[16]

In passing, we may observe that a complementary phenomenon to super-clusters is the rising tide of temporary agglomeration phenomena formed by the episodic build-up of externality effects leading to brief but intense congregations of many individuals in particular places. Examples are events such as business conventions, trade fairs, scientific congresses, and so on. In such cases, participants from widely scattered

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locations come together for short periods of time in order to engage in information exchange, discussion and debate, renewal of personal contact, and collective action.

The regional motors of the global economy can be likened to the theoretical super-clusters described above. As political barriers to economic interaction continue to fade, these motors may be expected to become, at least for a time, an overwhelmingly dominant element of the world system. I submit that we are poised at the early stages of the rise of a global economic order characterized by just such a set of regional motors as this. This claim is all the more emphatic because many of these motors are in practice made up of intersecting and overlapping (but distinctive) industrial sub-systems that usually help to boost the total local stock of agglomeration economies to yet higher levels than would obtain under conditions of strictly mono-industrial development. Although small and medium-sized clusters of economic activity thrive in huge numbers throughout the modern world, the current situation is one in which the incidence and size of super-clusters are also clearly on the increase.

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The Specific Theory of Locational Agglomeration

These last remarks call for further substantiation, and above all for further elaboration in terms of a specific theory of spatial agglomeration. The discussion now proceeds on two main fronts. First, I lay out some essentials of a purely static and transactional approach to understanding agglomeration. Second, I develop a story about the dynamics of agglomeration processes, paying particular attention to the path dependency and emergent effects of regional production complexes.

The static transactional theory

Any economic or social system can always be decomposed, at one level, into a dense criss-crossing network of transactional relationships. Such relationships are the essential foundation, even if they are not the totality, of all varieties of human interaction. They sustain the social division of labor, they are one of the bases of local labor markets, they function as the indispensable scaffolding of social and cultural routines, they are integral to the dissemination of many types of externality. And, precisely because they are spatially extensive and hence incur spatially dependent economic and social costs, they play a major role in the structuring of locational events.

At an earlier stage, I showed how the costs of transacting across geographic space vary enormously depending not just on distance, but also on the quantitative, technical, and social attributes of the transactional system. Wherever these costs are high, especially where there exist dense articulations of transactions in economic space, there is always some reasonable expectation of finding that the firms involved will be located in close proximity to one another in geographic space. We would in the first instance expect agglomeration to be fairly prominently developed in the case of highly disintegrated production systems, whose associated transactional networks are typified by many small transactions which are constantly changing in terms of their content and destinations, and which also require labor-intensive techniques of mediation. In addition, once proximity between producers is established, the rate of flow of circulating capital can be greatly accelerated, and the production schedules of interrelated firms can be more finely calibrated. In the Japanese *kanban* system, for example, mutual proximity of participant firms is commonly (though by no means ↳ invariably) needed for the system to work effectively. Any agglomeration of the sort under discussion here, then, will almost always be constituted as a collection of specialized but complementary economic activities held together by labyrinthine, high-cost transactional networks, and above all reinforced by the many increasing-returns effects that come into being as agglomerations form.

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Labor is also, and of necessity, an element of any such system, and any tendency to agglomeration will be further enhanced by the local labor-market structures that always take shape around any center of production. This remark raises intricate questions that cannot in any sense be adequately dealt with in the present account, and only a brief sketch of some preliminary ideas will be attempted.[17] These ideas are all related to the initial observation that the daily journey to work represents a particularly expensive kind of transaction per unit of distance so that any given set of employment places—if they are to be viable—must always co-exist in close spatial association with a set of workers' residential locations.

Three main analytical points flow from this observation. First, whenever employment places and workers' residences converge spatially, processes of job search, recruitment, and matching are facilitated because any given quantum of labor-market information can be circulated that much more easily among the relevant parties. Second, by dint of shared patterns of habituation, the work-force as a whole will come to be imbued with skills and aptitudes attuned to agglomeration-specific needs. Third, with the assemblage of a large work-force in one place, it becomes feasible to provide multifaceted educational and training programs responsive to the requirements of the local production system.

All of these local labor-market features are strongly interlaced with increasing-returns effects so that the larger the agglomeration, the more efficient its operation will tend to be. In the same way, the *per capita* costs of providing infrastructure artifacts and other public goods with high fixed costs and productivity-boosting virtues (such as expressways, airports, street-cleaning services, and so on) will usually be lower in large agglomerations than in small.

p. 93 One final major point about the transactional foundations of agglomeration needs to be made. Transactions are a crucial element of those beneficial 'atmospheric' effects that are indispensable if production complexes are to emerge not just as amalgams of capital and labor, but also as communities and as systems of untraded interdependencies; and again, their efficacy is enhanced by mutual proximity of the various participants. Proximity makes possible frequent personalized encounters between many different individuals occupying many different kinds of socio-economic niches, and thus positively encourages the evolution of distinctive local cultures. These cultures facilitate economically useful processes of socialization, and they provide an important context of mutual understandings within which learning and innovation can proceed at an accelerated rate (though they can also sometimes be a hindrance to development by placing barriers in the way of advantageous change). Learning and innovation are further enhanced by the constant interchanges of information that occur in industrial agglomerations.[18] The general idea is felicitously captured in the following account of traditional waggon-building in rural England at the turn of the century, an account in which the socialized nature of the knowledge base of the craft is clearly expressed:

> The lore was a tangled network of country prejudices, whose reasons were known in some respects here, in others there, and so on. In farm-yard, in tap-room, at market, the details were discussed over and over again; they were gathered together for remembrance in village workshops; carters, smiths, farmers, wheel-makers, in thousands handed on each his own little bit of understanding, passing it to his son or to the wheel-wright of the day, linking up the centuries. But for the most part, the details were but dimly understood; the whole body of knowledge was a mystery, a piece of folk-knowledge residing in the folk collectively but never wholly in any individual.[19]

Despite the reference to country prejudices, the same essential processes of social exchange and information circulation are critical to the operation of modern industrial regions, even such high-technology industrial regions as Orange County, California, with its advanced electronics manufacturers, or San Diego, with its burgeoning biotechnology industry. Regions where this sort of interchange occurs on a major scale are often p. 94 hotbeds of creativity, as illustrated by the streams ↳ of technological innovations that characterized the cotton textile industry in nineteenth-century Lancashire, or that are discernible in the semiconductor industry in Silicon Valley today. This endemic creativity of at least some agglomerations often endows them with absolute

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competitive advantages in production, enabling them to command global markets so long as they stay at the forefront of innovation.

Even in a purely static world, then, agglomeration is a forceful mechanism for begetting and appropriating many economic and social benefits. The dynamic aspects of agglomerations greatly magnify these properties.

The dynamic theory

Romer has shown that situations where increasing returns effects prevail are characterized by a temporal logic in which growth leads constantly onward to yet more growth.[20] Concomitantly, in any given regional cluster at any given instant, we are unlikely to detect anything even approaching static equilibrium. What we are much more sure to observe is a cross-section through a developmental trajectory that can be understood only in terms of a path-dependent process of evolution and adjustment structured by the phenomenon of agglomeration economies.[21]

At the outset, we need to relate the preceding remarks to a basic description of growth and change in articulated economic spaces and derivative agglomerations. I have already indicated that quantitative increases in the market demand for any particular product often lead to the formation of new and specialized tasks or trades in production systems, and these will constitute independent sectors if vertical disintegration also comes about. The process is accentuated where we have flexible production systems facing much uncertainty and instability so that frequent adjustments of process and product configurations occur, as in the case of heavily urbanized sectors like high-technology electronics, craft manufacturing (e.g. clothing), cultural-products industries, or business services. Thus, as markets expand, we often observe a corresponding pattern p. 95 of industrial change as ↳ depicted in Figure 5.3. What is shown here in very schematic outline is an industrial complex undergoing transformation both in the horizontal dimension (i.e. increases in the number of production establishments of any given type) and in the vertical dimension (i.e. increases in the number of different sectors as a result of dynamic vertical disintegration). To be sure, this pattern of change can always be undercut at any time by technological innovations that provoke radical resyntheses of the social division of labor, but so long as it continues it will exhibit the processes of division, diversification, multiplication, and specialization as described. For the sake of argument, I shall assume that the complex depicted in Figure 5.3 is endowed with strong static agglomeration economies at all times throughout its evolution.

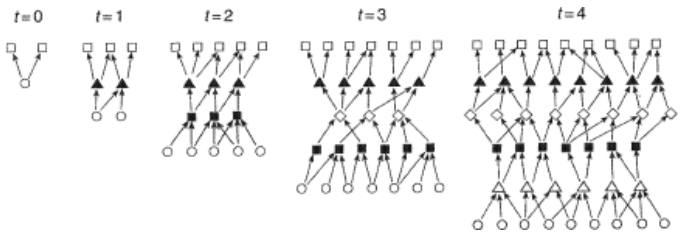

Fig. 5.3. A growing industrial complex with dynamic vertical disintegration, showing evolution of the system from three establishments in two sectors at t = 0 to forty-seven establishments in six sectors at t = 4

I shall further take it for granted that this complex, as it begins to make its appearance on the economic landscape, is indifferent to existing macro-geographic arrangements, and that It is viable—initially at least—in a wide variety of locations. Hence, this imaginary complex in its infant stage is not too dissimilar from a number of familiar empirical cases, such as cars in the 1890s, aircraft assembly in the early decades of the present century, or semiconductors in the 1950s. This means that the locational structure of the infant industry will be largely indeterminate from an economic point of view. Its early geography can be seen simply as an 'accident'—an effect, for example, of where its founding figure(s) happened to be living, or a result of a peculiar constellation of political forces in a particular place at a particular time.

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Let us suppose for the sake of argument that this structure comprises several different locales, no one of which has any special pre-given advantage over the others. Even so, small chance events alone are likely to push one locale into a leading position, if only in the sense that it begins to expand more rapidly than the others.[22] In other instances, a given locale may experience In the post-infant industry stage what we might designate as a 'breakthrough moment', namely, a decisive commercial or technological incident that pushes it to the forefront of development. Examples of this phenomenon are Henry Ford's organizational and managerial experiments in Detroit, Donald Douglas's development of the DC-3 aircraft in Los Angeles, and, arguably, the establishment of
p. 96 the Fairchild Semiconductor plant (with its Internal managerial dysfunctions and subsequent spin-offs) in ↳

p. 97 ↳
Silicon Valley. Once this occurs, there is a good chance that the locale will start to consolidate and extend its lead, especially where increasing returns and dynamic learning effects come into play. Because of lock-in effects, an early start is in fact often critical to long-run success in these kinds of system.

Provided that markets continue to grow, the leading locale is now likely to be subject to a many-sided process of developmental self-transformation in which the agglomeration processes described earlier will be greatly amplified, There is apt to be further elaboration of the social division of labor leading to economic diversification and increased economic synergies over a limited area. Concomitantly, local labor markets will grow, and new labor skills will almost certainly emerge. The industrial atmosphere of the locale will tend to become more finely delineated, and the business community may well begin to acquire identifiable cultural attributes marked by distinctive conventions and routines. Information exchanges and learning effects will become increasingly densely textured, with a corresponding sharpening of the stimuli to innovation. The ramifying social division of labor will in turn throw out more and more commercial opportunities, thus stimulating new entrepreneurial efforts. As these processes move forward, a complex regional economic system will progressively consolidate its geographic presence, and—at least for a time—will evolve forward on the basis of its deepening stock of external economies of scale and scope.

There are always, of course, numerous hazards (including the onset of agglomeration diseconomies) scattered along this pathway of development, and things do not always work out in practice quite so unproblematically, as illustrated by the cases of regional collapse that occur from time to time. However, once a region has made a good head start, so that increasing returns are being generated with some intensity and its competitive advantages are being continually sharpened, it will tend over some fairly long-run period to follow a recursive growth trajectory, providing that markets for its outputs remain healthy. This kind of growth is typically associated with lock-in, in the sense that many indurated and mutually reinforcing relationships within the regional economy will ensure that its trajectory acquires a marked dependence on its past.[23] This does not
p. 98 signify that the region ↳ will now be set irrevocably on a course to one final historical destination. To the contrary, even though lock-in implies that the region's development is path-dependent, it is also consistent with the existence of many branching-points representing conjunctures where the regional economy may move in any one of a number of different possible directions. Once it has moved, however, its future is then to that degree committed.

Unless there are bounds to the continued appropriation of increasing returns effects by producers in the region, development will continue in this manner, and there is some likelihood that sooner or later the region will become a pioneering center of development in its specialized domain of economic activity. Thus, what began as a footloose infant industry, will now—in this ideal-typical world—have attained a stage of geographical and historical development where it can be effectively carried on only in a limited number of locational contexts, i.e. those that provide just the right kinds of agglomeration economies and competitive advantages. By the same token, regions that fail to make an early start in fostering the development of a particular industry, or that fall behind in some way, are susceptible to 'lock-out' in the sense that they are liable to find it increasingly difficult to catch up with—much less overtake—the leading contenders.

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A Historical Geographic Perspective

A differentiated system of production agglomerations has always been an essential feature of capitalism, and in one way or another these agglomerations can generally be understood in terms of the abstracted principles outlined above. In practice, however, these principles always operate in unique geographical and historical contexts, resulting in very different kinds of contingent effects from case to case. Also, and quite apart from the repercussions that variables such as sector, scale of production, market conditions, or nation may have on the way these principles function, they are always subject to massive recomposition by reason of periodic shifts in the modalities of accumulation in capitalism.

Three brief examples may be advanced to illustrate the importance of the interplay between the systematic and the contingent in molding actual patterns of regional development at distinctively different periods of economic history. Equally, they serve to re-emphasize the idea of the present conjuncture of historical
p. 99 geography as a nexus of ↳ socio-economic relationships that will itself in due course be subject to fundamental restructuring.

The first example refers to the workshop and factory system that came so strongly to the fore in Britain in the first half of the nineteenth century, and that gave rise to a veritable revolution in patterns of regional development at that time. Major segments of the industrial apparatus in early nineteenth-century Britain were made up of small and vertically disintegrated firms at the core of dense transactional networks. The geography of production was accordingly, and to a major extent, arranged in classical Marshallian industrial districts forming much of the economic base of the manufacturing towns that characterized the period. Familiar examples are cottons in Lancashire, woollens in Yorkshire, cutlery in Sheffield, and the various metal trades of Birmingham. An impoverished urban working class keeping these industries going was constantly replenished by a massive drift of population from the countryside to the city—a contingent but critical condition of survival for the entire system.

Another example is offered by the system of fordist mass production that flourished in the United States from the 1920s to the 1960s, and which also brought about significant transformations of the economic landscape. In this instance, and to reiterate some of the information already presented in Chapter 2, the leading edges of production were to a great degree embodied in the lead plants of growth pole industries around which multi-tiered complexes of suppliers congealed in both functional and spatial terms. Such complexes typically constituted the foundations of the overgrown industrial metropolitan regions of the twentieth century. They were also the fountainhead of the blue-collar/white-collar social structures that characterized the cities of the US Manufacturing Belt over this period, and that were then re-expressed in distinctive patterns of socially segmented neighborhood development.[24] By the end of the 1970s, the competitive energies of fordist mass production were all but exhausted in most of the advanced capitalist countries, leading to a severe and prolonged crisis of the metropolitan areas that had grown up in response to its developmental thrust over the previous half-century.

A third and last example is presented by contemporary processes of post-fordist industrialization. This third
p. 100 case is distinguished by a ↳ multiplicity of flexible production networks in industries as diverse as fashion clothing, furniture, electronic components, biotechnology, films, and financial services, and concomitantly by a resurgence of industrial districts and agglomerations across the globe, even in areas located in what was formerly widely viewed as a development-resistant periphery. In their turn, the regions that have most actively participated in this type of industrialization now also find themselves bound tightly together in world-wide webs of interdependence, with multinational firms playing a major role in mediating between the local and the global. As a corollary, and in contrast to the older centralized multinationals of High Fordism, these newer global firms are often quite fragmented in terms of their command structure and functional organization. The

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privileged foci of this curent model of industrialization and urban development are the big regional motors of the new global economy, especially in North America, Western Europe, and East and South-east Asia.

This latest episode in the historical geography of capitalism is based on production systems that tend to be quite fragmented and diversified, and yet which are also capable of efficient operation at many different levels of scale, individually or collectively.[25] We might say—very schematically—that, whereas nineteenth-century workshop and factory systems were able to produce variety of output but were limited in the total scale that they could achieve, and whereas fordist mass production freed industry from quantitative restraint but at the expense of product variety, modern flexible production systems (with the aid of new electronic and information technologies) are able to achieve considerable variety of output while they can also often attain significant scale effects. Further, because post-fordist, flexible production systems are generally strongly externalized and transactions-intensive, regional agglomeration seems to be again resurgent, in contrast with the steady break-up of many industrial regions that was occurring as fordism approached its climacteric. Regions are once more emerging as foci of production and repositories of specialized know-how and technological capability, even as the globalization of economic relationships proceeds at an accelerating rate.

Notes

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